Seonghwan Park

Researcher, Korea Electronics Technology Institute (KETI)

shpark97@keti.re.kr
shpark97.github.io

About

I am a researcher at the Korea Electronics Technology Institute (KETI), working with Dr. Min-Gyu Park on **3D vision and physical AI**. I received my M.S. in Artificial Intelligence from POSTECH under Prof. Namhoon Lee, where I studied **optimization** for deep learning.

I am broadly interested in **efficiency and robustness in vision and machine learning**, which I approach through optimization: how a model reaches its optimum sets the cost of learning, and which optimum it reaches sets how well it generalizes. My work spans derivative-free optimization for black-box models and loss-landscape flatness for robustness to noisy labels and efficient OOD detection. I now carry this lens into 3D vision, reconstructing hand-object interactions, and hope to bring it to robotics grounded in vision priors.

Education

02/2023 – 08/2025 **Pohang University of Science and Technology (POSTECH), Korea**
M.S. in Artificial Intelligence, advised by Prof. Namhoon Lee
- GPA 3.9 / 4.0 (4.0 / 4.3)

03/2017 – 02/2023 **Hanyang University, Korea**
B.S. in Computer Science
- GPA 3.82 / 4.0 (4.12 / 4.5)
- Graduated *summa cum laude*
- Leave of absence for military service (10/2018 – 08/2020)

Publications

* [P] In preparation | [C] Conference | [W] Workshop

[P1] Towards Unified and Generalizable Monocular 3D Hand-Object Interaction Reconstruction
Seonghwan Park, Yong-Hoon Kwon, Min-Gyu Park
Ongoing project

[C5] Recovering Black-Box Point Cloud Models from Pose Shifts via Derivative-Free Optimization [Paper]
Seonghwan Park, Min-Gyu Park
*Winter Conference on Applications of Computer Vision (**WACV**), **2027** • Accepted as a Round 1 paper*

[C4] Exploiting Local Flatness for Efficient Out-of-Distribution Detection [Paper]
Seonghwan Park, Hyunji Jung, Dongyeop Lee, Namhoon Lee
*European Conference on Computer Vision (**ECCV**), **2026***

[C3] JacobianAvatar: Temporally Consistent Semi-rigid Avatar Reconstruction from a Monocular Video [Paper]
Changyeon Won, Min-Gyu Park, Seonghwan Park, Ju Hong Yoon, Hae-Gon Jeon
*European Conference on Computer Vision (**ECCV**), **2026***

[C2] An Analysis of Concept Bottleneck Models: Measuring, Understanding, and Mitigating the Impact of Noisy Annotations [Paper]
Seonghwan Park, Jueun Mun, Donghyun Oh, Namhoon Lee
*Neural Information Processing Systems (**NeurIPS**), **2025***

[C1] ZIP: An Efficient Zeroth-order Prompt Tuning for Black-box Vision-Language Models [Paper]
Seonghwan Park, Jaehyeon Jeong, Yongjun Kim, Jaeho Lee, Namhoon Lee
*International Conference on Learning Representations (**ICLR**), **2025***

[W2] FedFwd: Federated Learning without Backpropagation [Paper]
Seonghwan Park, Dahun Shin, Jinseok Chung, Namhoon Lee
*Workshop on Federated Learning and Analytics in Practice @ ICML (**ICML Workshop**), **2023***

[W1] On the Effectiveness of Sharpness-Aware Minimization with Large Mini-batches [Paper]
Jinseok Chung, Seonghwan Park, Jaeho Lee, Namhoon Lee
*Workshop on High-dimensional Learning Dynamics @ ICML (**ICML Workshop**), **2023***

Research Experience

01/2026 – Present **Korea Electronics Technology Institute (KETI)**
Researcher (Supervisor: Dr. Min-Gyu Park)
- Research Area: 3D vision and physical AI ([P1], [C3], [C5])

01/2023 – 12/2025 **Computational Optimization Lab, POSTECH**
Research Assistant (Supervisor: Prof. Namhoon Lee)
- Research Area: Model efficiency and robustness through optimization ([C1], [C2], [C4], [W1], [W2], [PT1]–[PT3])

06/2025 – 11/2025 **Samsung Research Project** ($70,000 USD/year)
- Research Area: Multi-modal foundation models for anomaly detection

06/2024 – 05/2025 **Samsung Advanced Institute of Technology (SAIT) Project** ($70,000 USD/year)
- Research Area: Visual prompting for MMFM-based visual question answering ([C1], [PT3])

01/2022 – 12/2022 **Machine Learning Lab, Hanyang University**
Undergraduate Research Intern (Supervisor: Prof. Eun-Sol Kim)
- Research Area: Prompt tuning for LLM-based VQA

Honors & Awards

2025 **NeurIPS Travel Grants** (Travel support for NeurIPS 2025, $1,925 USD)
2025 **CKAIA Best Paper Finalist Award** ($360 USD)
2022 **HYU Academic Excellence Award** (Top 3%)
2022 **Merit-based Scholarship** (1 semester, $1,300 USD), Hanyang University
2017 – 2018 **Career-based Scholarship** (4 semesters, $13,000 USD), Hanyang University

Patents

[PT3] A Method for Generating Data, Obtaining Data and an Electronic Apparatus Performing the Same
Hyunjeong Lee, Seonghwan Park, Namhoon Lee, Jaehyeon Jeong, Seungin Park
Under registration review, South Korea

[PT2] Electronic Device for Training Artificial Intelligence Model in Black Box Environment by Using Prompts and Operating Method of the Same
Namhoon Lee, Jaeho Lee, Seonghwan Park, Jaehyeon Jeong, Yongjun Kim
Under registration review, US & South Korea

[PT1] Apparatus and Method for Reducing the Impact of Label Noise on the Prediction Accuracy of a Conceptual Bottleneck Model
Namhoon Lee, Seonghwan Park, Jueun Mun
Under registration review, US & South Korea

Teaching

09/2024 – 12/2024 **Teaching Assistant** — POSTECH, *Introduction to Artificial Intelligence*
05/2023 – 06/2023 **Teaching Assistant** — POSCO AI Fellowship, *Deep Learning*

Services

2026 – 2027 **Paper Reviewing** — ICLR 2026, ECCV 2026, NeurIPS 2026, WACV 2027, TMLR
10/2018 – 08/2020 **Military Service** — Sergeant, Republic of Korea Air Force (ROKAF) Logistics Command

Skills & Languages

Programming Languages Python, C/C++, C#, PyTorch, LaTeX, Git
Korean (native), English (professional working)

References

Prof. Namhoon Lee *(M.S. Advisor)*
Associate Professor
Pohang University of Science and Technology (POSTECH)
namhoon.lee@postech.ac.kr

Dr. Min-Gyu Park *(Current Supervisor)*
Principal Researcher
Korea Electronics Technology Institute (KETI)
mpark@keti.re.kr